Package

Create a package which contains five modules fish, birds, amphibians, mammals, and reptiles. Each module contains two functions example and characteristics.

```
amphibians.py
def examples():
    print("Here are some examples of amphibians:")
    egs=['Frog',"Salamander","Toads","Newts","caecilians"]
    for f in egs:
        print(f)
def chars():
    ch=["Cold blooded","Lays eggs","Moist scaleless skin"]
    print("Characteristics of amphibians:")
    for c in ch:
        print(c)

birds.py
def examples():
    print("Here are some examples of birds:")
    egs=['Parrot',"Pigeon","Crow","Owl","Sparrow"]
    for f in egs:
        print(f)
def chars():
    ch=["Have wings","Lays eggs","Warm blooded","Have beaks and no teeth"]
    print("Characteristics of birds:")
    for c in ch:
        print(c)

fish.py
def examples():
    print("Here are some examples of fish:")
    egs=['Goldfish',"Tuna","Guppy","Piranha","Swordfish"]
    for f in egs:
        print(f)
def chars():
    ch=["Lives Under Water","Breathes through gills","Swims using fins and Tail"]
    print("Characteristics of fish:")
    for c in ch:
        print(c)

mammals.py
def examples():
    print("Here are some examples of mammals:")
    egs=["Human","Cat","Tiger","Dog","Elephant"]
    for f in egs:
```

```
        print(f)
def chars():
    ch=["Warm blooded","Gives birth to babies","Have hair and fur","Four chambered heart"]
    print("Characteristics of mammals:")
    for c in ch:
        print(c)
```

reptiles.py

```
def examples():
    print("Here are some examples of reptiles:")
    egs=['Turtle',"Lizard","Crocodiles","Chameleon","Snakes"]
    for f in egs:
        print(f)
def chars():
    ch=["Have scales","Vertebrates","Breathe through lungs","Cold blooded"]
    print("Characteristics of reptiles:")
    for c in ch:
        print(c)
```

Creatures.py

```
import fish,birds,amphibians,mammals,reptiles
def __init__(self):
    print("Find your information about creatures:examples and characteristics")
```

main.py

```
#Creatures package includes various modules
import Creatures
print("Find your information about creatures:examples and characteristics")
while(1):
    print("Press\n1 for fish\n2 for birds\n3 for amphibians\n4 for mammals\n5 for reptiles\n6 to exit:")
    c=int(input())
    if c==1:
        Creatures. fish.examples()
        Creatures. fish.chars()
    elif c==2:
        Creatures. birds.examples()
        Creatures. birds.chars()
    elif c==3:
        Creatures. amphibians.examples()
        Creatures. amphibians.chars()
    elif c==4:
        Creatures. mammals.examples()
        Creatures. mammals.chars()
    elif c==5:
        Creatures. reptiles.examples()
        Creatures. reptiles.chars()
    elif c==6:
        print("Thank You!!!")
```

```
        break
    else:
        print("Wrong input")
```

#for each module class concept has been introduced purposefully.
#to be noted same process can be implemented without introducing separate class

File

Write a file. Read the file and print its content. Read first two lines. Read first 5 characters. Read the file using loop.
demofile.txt
Hello! Welcome to demofile.txt
This file is for testing purposes.
Good Luck!

```
f = open("demofile.txt", "r")
print(f.read())
f.seek(0)
print(f.readline())
print(f.readline())
f.seek(0)
print(f.read(5))
f.seek(0)
for x in f:
  print(x)
f.close()
```

Append content in the above mentioned file.

```
f = open("demofile2.txt", "a")
f.write("Now the file has more content!")
f.close()
#open and read the file after the appending:
f = open("demofile2.txt", "r")
print(f.read())
```

Check whether a file exists or not. Delete it.

```
        import os
if os.path.exists("demofile.txt"):
  os.remove("demofile.txt")
else:
  print("The file does not exist")
```

Count the number of words in a file.

```
words=0
with open(“demo.txt”,’r’) as f:
      for line in f:
         w=line.split()
         words+=len(words)
max_len=len(max(w,key=len))
print(words)
```

Plotting

Write a program to plot scattered points.

```
import matplotlib.pyplot as plt
def draw_multiple_points():
    x_number_list = [1, 4, 9, 16, 25]
    y_number_list = [1, 2, 3, 4, 5]
    plt.scatter(x_number_list, y_number_list, s=10)
    plt.title("Extract Number Root ")
    plt.xlabel("Number")
    plt.ylabel("Extract Root of Number")
    plt.show()
if __name__ == '__main__':
    draw_multiple_points()
```

Write a program to plot using lines.

```
import matplotlib.pyplot as plt
x=[1,2,3]
y=[2,4,1]
plt.plot(x,y,label='l1')
x=[1,2,3]
y=[8,6,9]
plt.plot(x,y,label='l2')
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.legend()
plt.show()
```